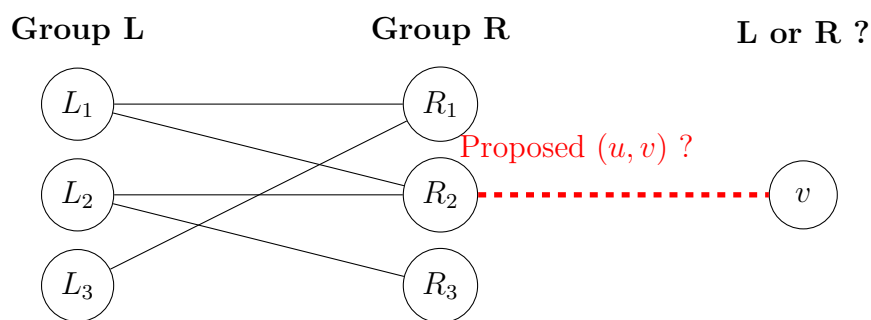


Student ID 1 & 2: (Do this in pairs)

1 The Network Integrity Challenge

Scenario: You are managing a communication network. The network is currently *Bipartite*, meaning the users are split into two groups (Group L and Group R), and users only communicate with members of the opposite group.



The Problem: You need to add one new edge between two existing users. However, if this edge creates an odd cycle, the bipartite property of the network will be destroyed.

The Task: Design a Verification Algorithm

1. **Algorithm Design:** Given an existing bipartite graph $G = (V, E)$ and a proposed new edge (u, v) , describe a simple strategy using BFS to determine whether adding (u, v) will break the bipartite property.

Hint: Think about the “layers” (or two-coloring/groups) assigned to nodes during a BFS traversal. Which group does each endpoint belong to?

2. **Efficiency:** What is the runtime of your algorithm? Can it be done in $O(V + E)$?